Physics 829: Homework Set No. 6

Due date: Wednesday, May 18, 2012, 1:00pm in PRB M2043 (Biao Huang's office)

Total point value of set: 70 points

Problem 1 (20 pts.):

Evaluate by contour integration the Fourier transform of

$$G_\epsilon(\boldsymbol{q}) = \frac{1}{(2\pi)^{3/2}} \frac{1}{k^2 - i\epsilon - q^2}$$

in the limit $\epsilon \to 0^+$. (Make sure to provide mathematically rigorous arguments for each step of your evaluation.) Show by explicit differentiation that it is the Green function of the Helmholtz operator (i.e. it solves the Helmholtz equation for a point source) with *incoming spherical wave* boundary conditions.

Problem 2 (10 pts.): Exercise 19.5.3 (Shankar, p. 554)

Problem 3 (20 pts.): Exercise 19.5.5 (Shankar, p. 554)

Problem 4 (20 pts.): Exercise 19.5.6 (Shankar, p. 555)

829. Quantum Mechanics HW #6

Biao Huang

Problem 1.

$$G(\vec{q}) = \frac{1}{(2\pi)^{3/2}} \frac{1}{k^2 - i\varepsilon - q^2}$$

F.T. $$G(\vec{r}) = \int \frac{d^3\vec{q}}{(2\pi)^{3/2}} G(\vec{q})\, e^{i\vec{q}\cdot\vec{r}}$$

$$= \frac{2\pi}{8\pi^3} \cdot \int_0^\infty q^2 dq \int_{-1}^{1} d\cos\theta \, \frac{e^{iqr\cdot\cos\theta}}{k^2 - i\varepsilon - q^2}$$

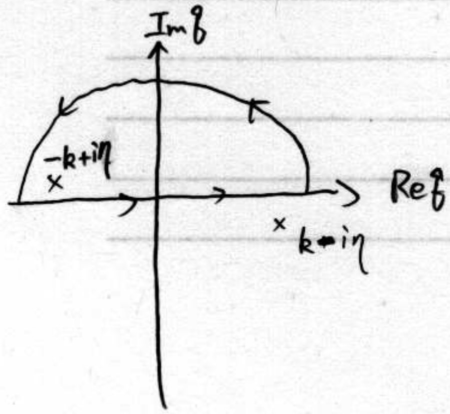
$$= \frac{1}{4\pi^2} \int_0^\infty q^2 dq \, \underbrace{\frac{1}{k^2 - i\varepsilon - q^2} \cdot \frac{1}{iqr} \cdot (e^{iqr} - e^{-iqr})}_{\text{even in } q.}$$

$$= \frac{1}{4\pi^2 ir}\left[\frac{1}{2}\int_{-\infty}^{\infty} \frac{q\,dq}{k^2 - i\varepsilon - q^2} \cdot (e^{iqr} - \underbrace{e^{-iqr}}_{q' = -q})\right]$$

$$= \frac{1}{8\pi^2 ir}\left[\int_{-\infty}^{\infty} \frac{q e^{iqr} dq}{k^2 - i\varepsilon - q^2} - \underline{\int_{\infty}^{-\infty} \frac{q' dq' \, e^{iq'r}}{k^2 - i\varepsilon - q'^2}}\right]$$

$$= -\frac{1}{4\pi^2 ir} \int_{-\infty}^{\infty} dq \, \frac{q e^{iqr}}{q^2 - k^2 + i\varepsilon}$$

Now, apply contour integration. ① Note $(e^{iqr})|_{q=-i\infty} \to \infty$, we need to enclose the contour on the upper plane.



② From Jordan's lemma, if the integrand

$$f(q) = \frac{q}{q^2-k^2+i\varepsilon}$$

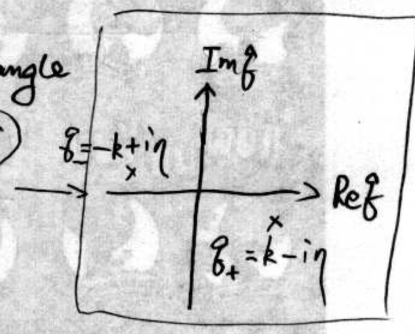
satisfies $(f(q)/q)\big|_{|q|\to\infty} = 0$,

The integration $\int_{arc} f(q) e^{iqr}\, dq$ along the infinite arc vanishes. Thus, the contour integral equals the original real integration.

③ The singularities for $\frac{1}{q^2-k^2+i\varepsilon}$ can be specified as follows.

$$q^2-k^2+i\varepsilon \overset{\varepsilon\to0}{\simeq} q^2-(k^2e^{-i\varepsilon+2n\pi i})$$

$$\Rightarrow q_\pm = \sqrt{k^2e^{-i\varepsilon+2n\pi i}} = ke^{\overbrace{-\frac{i\varepsilon}{2}+n\pi i}^{\text{complex angle}}} \overset{\eta=\frac{\varepsilon}{2}}{=} \pm(k-i\eta)$$



①②③ $\Rightarrow$ Apply residue theorem

$$\boxed{G(\vec{r})} = -\frac{1}{4\pi^2 ir}\cdot 2\pi i\left[\frac{qe^{iqr}}{(q-q_+)(q-q_-)}(q-q_-)\right]\Bigg|_{q=q_-,\ \eta\to0}$$

$$= -\frac{1}{2\pi r}\cdot\frac{-ke^{-ikr}}{-2k} = \boxed{-\frac{1}{4\pi}\frac{e^{-ikr}}{r}}$$

Differentiating it directly.

$$\nabla^2 G(\vec{r}) = -\frac{1}{4\pi}\nabla^2\Big[\underbrace{\Big(\frac{e^{-ikr}-1}{r}\Big)}_{\substack{\text{singularity free} \\ \text{at } r=0. \\ \Rightarrow \text{no } \delta(\vec{r}) \\ \Rightarrow \text{can apply } \nabla^2 \to \frac{1}{r}\frac{\partial^2}{\partial r^2} r}} + \underbrace{\frac{1}{r}}_{\nabla^2 \frac{1}{r} = -4\pi\delta(\vec{r})}\Big]$$

$$= -\frac{1}{4\pi}\Big[\frac{1}{r}\frac{\partial^2}{\partial r^2}(e^{-ikr}-1) + (-4\pi\delta(\vec{r}))\Big]$$

$$= \underbrace{-\frac{e^{-ikr}}{4\pi r}}_{G(\vec{r})}(-k^2) + \delta(\vec{r}) = -k^2 G(\vec{r}) + \delta(\vec{r})$$

$$\Rightarrow (\nabla^2 + k^2)G(\vec{r}) = \delta(\vec{r})$$

— which is desired.

Problem 2. (19.5.3. Shankar P 554)

① The general formula $\sigma_l = \frac{4\pi}{k^2}(2l+1)\sin^2\delta_l$

& the asymptonic behavior for δ_0 in the hard sphere case

$$\underset{(19.5.29)}{\delta_0 \xrightarrow{k\to 0} (kr_0)}$$

tells $\sigma_0 \xrightarrow{k\to 0} \frac{4\pi}{k^2}\cdot\delta_0^2 = \frac{4\pi}{k^2}\cdot(kr_0)^2 = 4\pi r_0^2$

② When $kr_0 \to \infty$,

$$\delta_l \underset{(19.5.27)}{\overset{kr_0\to\infty}{=}} \tan^{-1}\left(\frac{j_l(kr_0)}{n_l(kr_0)}\right) = \tan^{-1}\left(\frac{\sin(kr_0-\frac{l\pi}{2})}{kr_0}\Big/\left(-\frac{\cos(kr_0-\frac{l\pi}{2})}{kr_0}\right)\right)$$

$$= -\left(kr_0-\frac{l\pi}{2}\right), \text{ so } \sin^2\delta_l \xrightarrow{kr_0\to\infty} \sin^2\left(kr_0-\frac{l\pi}{2}\right)$$

$$\Rightarrow \sigma = \sum_{l=0}^{\infty}\frac{4\pi}{k^2}(2l+1)\sin^2\left(kr_0-\frac{l\pi}{2}\right)$$

$$\simeq \frac{4\pi}{k^2}\int_0^{kr_0}(2l)\underbrace{\langle\sin^2(kr_0-\frac{l\pi}{2})\rangle}_{\to\frac{1}{2}}\,dl$$

$$= \frac{4\pi}{k^2}\cdot\frac{(kr_0)^2}{2} = 2\pi r_0^2$$

Problem 3.
(19.5.5. P554 Shankar)

From (19.5.38), the element of scattering matrix

$$S_l(k) = \frac{\text{out-going wave amplitude}}{\text{in-coming wave amplitude}}$$

For elastic scattering (with no absorption), $|S_l(k)| = 1$.

so $S_l(k) \equiv e^{2i\delta_l}$. If there is absorption the out-going wave should have smaller amplitude than the in-coming wave. So the physics can be charaterized by modulating $S_l(k) \to \eta_l(k) e^{2i\delta_l}$, with

$\eta_l(k) < 1$ denoting the absorption; then $|S_l(k)|^2 < 1$.

More concretely we can calculate the probability density.
The probability current related to scattering expt is the radial component, and at $r \to \infty$ where observation points locate,

$$R_{kl}(r) \to A \frac{e^{ikr}}{r} + B \frac{e^{-ikr}}{r} \qquad (19.5.39)$$

& The whole wave function

$$\psi_{kl}(\vec{r}, \theta) = P_l(\cos\theta) R_{kl}(r)$$

Then we can apply the general formula for current density

$$\vec{j}_{kl} = \frac{1}{2}\left[\psi_{kl}^* \frac{\vec{p}}{m}\psi_{kl} - \psi_{kl}\frac{\vec{p}}{m}\psi_{kl}^*\right]$$

$$= -\frac{i\hbar}{2m}\Big[\underbrace{\psi_{kl}^* \vec{\nabla}\psi_{kl} - \psi_{kl}\vec{\nabla}\psi_{kl}^*}_{=2i\,\mathrm{Im}\,\psi^*\vec{\nabla}\psi}\Big]$$

$$= \frac{\hbar}{m}\,\mathrm{Im}\left(\psi_{kl}^*\vec{\nabla}\psi_{kl}\right).$$

Now, $\boxed{\vec{\nabla}\psi} = \Big(\vec{e}_\theta, \vec{e}_\varphi, \underbrace{\frac{\vec{e}_r}{r^2}}_{\text{higher order when } r\to\infty}$ components which are irrelevant$\Big)$

$$+ P_l(\cos\theta)\frac{\vec{e}_r}{r}\frac{\partial}{\partial r}\left(Ae^{ikr} + Be^{ikr}\right)$$

$$= \left(\vec{e}_\theta, \vec{e}_\varphi, \frac{\vec{e}_r}{r^2} \text{ components}\right)$$

$$+ P_l(\cos\theta)\cdot\frac{\vec{e}_r}{r}\, ik\left(Ae^{ikr} - Be^{-ikr}\right)$$

$$\Rightarrow (j_{kl})_r \overset{r\to\infty}{=} \frac{\hbar}{m}\,\mathrm{Im}\left[\left(P_l\,\frac{A^*e^{-ikr} + B^*e^{+ikr}}{r}\right)\cdot\left(P_l\,\frac{ik(Ae^{ikr} - Be^{-ikr})}{r}\right)\right]$$

$$= \frac{\hbar}{mr^2}|P_l|^2\,\mathrm{Im}\Big[ik\Big(|A|^2 - |B|^2 \underbrace{- A^*Be^{-2ikr} + B^*Ae^{2ikr}}_{\text{Purely imaginary}}\Big)\Big]$$

$$= \frac{\hbar k}{mr^2}|P_l|^2\left(|A|^2 - |B|^2\right)$$

From $S_l(k) = \frac{A}{B}$ in (Shankar 19.5.38),

$$\boxed{(j_{kl})_r = \frac{\hbar k}{mr^2}|P_l|^2\,|B|^2\left(|S_l(k)|^2 - 1\right)}$$

Thus, if $S_\ell(k) = e^{2i\delta_\ell}$, which is unitary, we have $(j_{k\ell})_r = 0$ at $r \to \infty$ which implies conservation of probability:

$$\oint (j_{k\ell})_r \, d\Omega = -\frac{\partial}{\partial t}\int \rho \, dV = \underbrace{0}_{\text{no particle lose in the whole space}}$$

If there is absorption $(j_{k\ell})_r < 0$, and therefore

$$S_\ell(k) = \eta_\ell(k)\, e^{2i\delta_\ell},$$

modulated by a factor $\eta_\ell(k)$.

① The cross section is given by

$$\sigma_{el} = \int |f(\theta)|^2 \, d\Omega$$

where $f(\theta) = \sum_\ell \frac{S_\ell(k) - 1}{2ik}(2\ell+1) P_\ell(\cos\theta)$ according to (19.3.13-15). Then from $\int P_\ell P_{\ell'} \, d\Omega = \frac{2 \times 2\pi}{2\ell+1}\delta_{\ell\ell'}$ $\quad \int_0^{2\pi} d\phi$

$$\Rightarrow \sigma_{el} = \sum_\ell \frac{|S_\ell(k) - 1|^2}{4k^2}(2\ell+1)^2 \frac{4\pi}{2\ell+1}$$

$$= \frac{\pi}{k^2}\sum_{\ell=0}^{\infty}(2\ell+1)|S_\ell(k) - 1|^2$$

Using $S_\ell(k) = \eta_\ell(k)\, e^{2i\delta_\ell}$, we have

$$|S_\ell(k) - 1|^2 = |\eta_\ell(k)|^2 + 1 - 2\eta_\ell(k)\cos 2\delta_\ell$$

$$\Rightarrow \boxed{\sigma_{el} = \frac{\pi}{k^2}\sum_{\ell=0}^{\infty}(2\ell+1)\left(1+\eta_\ell^2 - 2\eta_\ell\cos 2\delta_\ell\right)}$$

The inelastic cross section is

$$\frac{d\sigma_{inel}}{d\Omega} = \left|\frac{j_r}{j_i}\right| \cdot r^2 \quad , \quad \sigma_{inel} = \int \left|\frac{j_r}{J_i}\right| \cdot r^2\, d\Omega \, .$$

From page 6 we have $(j_{k\ell})_r = \frac{\hbar k}{m r^2} |P_\ell|^2 |B_\ell|^2 \left(|S_\ell(k)|^2 - 1\right)$

Note $B \sim$ incoming wave. and from

$$\psi = e^{ikz} + f(\theta)\underbrace{\frac{e^{ikr}}{r}}_{\text{out-going}}$$

we know it purely comes from (19.5-8)

$$e^{ikz} = \sum_\ell \frac{2\ell+1}{2ik} P_\ell(\cos\theta)\left[\frac{e^{ikr}}{r} - \underbrace{\frac{e^{-i(kr-\ell\pi)}}{r}}_{\text{here}}\right].$$

Compare with the form

$$\psi = \sum_\ell P_\ell(\cos\theta)\left[A_\ell \frac{e^{ikr}}{r} + B_\ell \frac{e^{-ikr}}{r}\right]$$

$$\Rightarrow B_\ell = \frac{2\ell+1}{2ik} e^{i\ell\pi}$$

Then, $\boxed{j_r} = \sum_{l=0}^{\infty} (j_{kl})_r = \sum_{l=0}^{\infty} \frac{\hbar k}{mr^2} |P_l|^2 |B_l|^2 (|S_l(k)|^2 - 1)$

$$= \frac{\hbar k}{mr^2} \sum_{l=0}^{\infty} P_l^2 \cdot \frac{(2l+1)^2}{4k^2} (\eta_l(k)^2 - 1) < 0$$

representing the incoming wave has higher current density than out-going one.

And $\boxed{j_i = \frac{\hbar k}{m}}$ (19.2.18) for $\psi_{inc} = e^{ikz}$

$$\Rightarrow \boxed{\sigma_{inel}} = \int \frac{\frac{\hbar k}{mr^2} \sum_{l=0}^{\infty} P_l^2(\cos\theta) \frac{(2l+1)^2}{4k^2} |\eta_l^2 - 1|}{\frac{\hbar k}{m}} r^2 d\Omega$$

$$= \sum_{l=0}^{\infty} \frac{4\pi}{2l+1} \cdot \frac{(2l+1)^2}{4k^2} (1 - \eta_l^2)$$

$$\boxed{= \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)(1-\eta_l^2)}$$

$$\Rightarrow \boxed{\sigma_{tot}} = \sigma_{inel} + \sigma_{el} = \boxed{\frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)(2 - 2\eta_l \cos 2\delta_l)}$$

$$\& \quad \frac{4\pi}{k} \operatorname{Im} f(\theta=0) = \frac{4\pi}{k} \sum_{l=0}^{\infty} \operatorname{Im} \left[\frac{S_l(k) - 1}{2ik} (2l+1) \underbrace{P_l(\cos\theta = 1)}_{=1} \right]$$

$$= \frac{4\pi}{k} \sum_{l=0}^{\infty} (-1) \frac{\eta_l \cos 2\delta_l - 1}{2k} (2l+1)$$

$$= \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)(2 - 2\eta_l \cos 2\delta_l)$$

$= \sigma_{tot}$, the optical theorem holds.

② "Black disk" : $\eta_l = \begin{cases} 0 & l \le kr_0 \\ 1 & l > kr_0 \end{cases}$.

$$\sigma_{el} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)(1+\eta_l^2 - 2\eta_l \cos 2\delta_l)$$

$$\simeq \frac{\pi}{k^2} \int_0^{kr_0} dl \; (2l+1) = \frac{\pi}{k^2}\left((kr_0)^2 + kr_0\right)$$

$$\overset{kr_0 \gg 1}{\cong} \frac{\pi}{k^2} \cdot k^2 r_0^2 = \pi r_0^2$$

$$\sigma_{inel} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)(1-\eta_l^2)$$

$$\cong \frac{\pi}{k^2} \int_0^{kr_0} (2l+1)\, dl \cong \frac{\pi}{k^2} (kr_0)^2 = \pi r_0^2$$

$$\Rightarrow \quad \sigma_{el} = \sigma_{inel} \simeq \pi r_0^2$$

Here σ_{el} ~ the portion of incident wave being "moved" (scatterred)

σ_{inel} ~ the portion of incident wave being absorped.

For a "black disc", all wave being scatterred because they necessarily go inside $r<r_0$, are absorbed. So "being scattered" = "being absorbed", and

$\sigma_{el} = \sigma_{inel}$

Problem 4 (19.5.6 Shankar P555)

— Optical Theorem

① $\psi = e^{ikz} + f(\theta)\frac{e^{ikr}}{r} = e^{ikr\cos\theta} + f(\theta)\frac{e^{ikr}}{r}$

$$\vec{j} = \frac{\hbar}{m}\,\mathrm{Im}(\psi^* \vec{\nabla}\psi)$$

$$= (\vec{e}_\theta, \frac{\vec{e}_r}{r^2}, \frac{\vec{e}_r}{r^3} \text{ components})$$

$$+ \vec{e}_r \cdot \frac{\hbar}{m}\,\mathrm{Im}\Big[\Big(e^{-ikr\cos\theta} + f^*(\theta)\frac{e^{-ikr}}{r}\Big) \times \Big(ik\cos\theta\, e^{ikr\cos\theta} + \frac{f(\theta)}{r}\cdot ike^{ikr}\Big)\Big]$$

$$= (\vec{e}_\theta, \frac{\vec{e}_r}{r^2} \text{ components})$$

Not interference

$$+ \vec{e}_r \frac{\hbar k}{m}\,\mathrm{Im}\Big[i\cos\theta + \frac{|f(\theta)|^2}{r^2}\, i$$

also higher order

$$+ \frac{ie^{ikr(\cos\theta-1)}}{r} f^*(\theta)\cos\theta + \frac{ie^{ikr(1-\cos\theta)}}{r} f(\theta)\Big]$$

Thus, the radial current due to interference of inc & sc waves, up to leading order, is

$$j_r \xrightarrow{r\to\infty} \frac{\hbar k}{m}\frac{1}{r}\,\mathrm{Im}\Big[ie^{ikr(\cos\theta-1)} f^*(\theta)\cos\theta + ie^{ikr(1-\cos\theta)} f(\theta)\Big]$$

— as desired

② When $\theta \neq 0$, $e^{\pm ikr(\cos\theta - 1)}$ oscillates fast with the change of θ, while other pieces $f(0)$, $f^*(0)$, $\cos\theta$ remains well behaved. Thus, the accumulative effect is that different phases $e^{\pm ikr(\cos\theta-1)}$ cancel each other, resulting in 0.

③ When $\theta \to 0$, $\cos\theta - 1 \to 0$. We can expand

$$\cos\theta - 1 \simeq (1 - \frac{\theta^2}{2}) - 1 = -\frac{\theta^2}{2}$$

$$\Rightarrow \int_{corn} j_r\, r^2 d\Omega \simeq \frac{\hbar k}{m} \frac{1}{r} \int_{corn} r^2 d\Omega \; \mathrm{Im}\Big[i \underbrace{e^{+ikr(\cos\theta-1)}}_{average} f^*(0) + i \underbrace{e^{ikr(1-\cos\theta)}}_{average} f(0) \Big]$$

$$\simeq \frac{\hbar k}{m} r \underset{\uparrow \atop \int_0^{2\pi} d\varphi}{2\pi} \cdot \mathrm{Im}\Big[i \frac{f(0)^*}{ikr} - i \frac{f(0)}{ikr} \Big]$$

$$= \frac{\hbar k}{m} \frac{2\pi}{k} \cdot \underbrace{\mathrm{Im}[f^*(0) - f(0)]}_{= -2\,\mathrm{Im} f(0)}$$

$$= -\left(\frac{\hbar k}{m}\right) \frac{4\pi}{k} \mathrm{Im} f(0)$$

Since $j_{inc} = \frac{\hbar k}{m}$, $\sigma = \int \left|\frac{j_r}{j_{inc}}\right| r^2 d\Omega = \frac{4\pi}{k} \mathrm{Im} f(0)$